

Hardware replacement tips

Replacing parts

Lenovo

Screw types

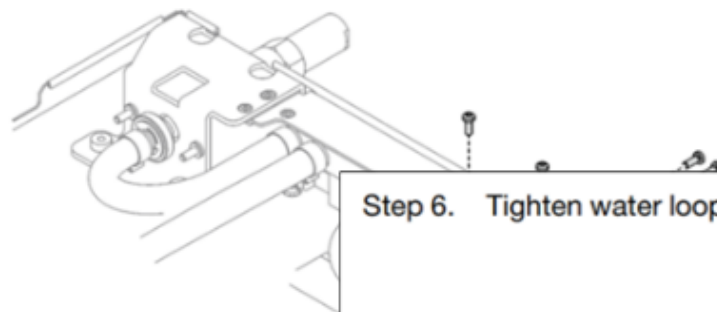
Prepare the following screwdrivers for SD650 V3, SD650-N V3, or SD650-I V3 part replacement:

- Torx T10 screwdriver
- Torx T15 screwdriver
- Torx T30 screwdriver
- Phillips #1 screwdriver
- Phillips #2 screwdriver

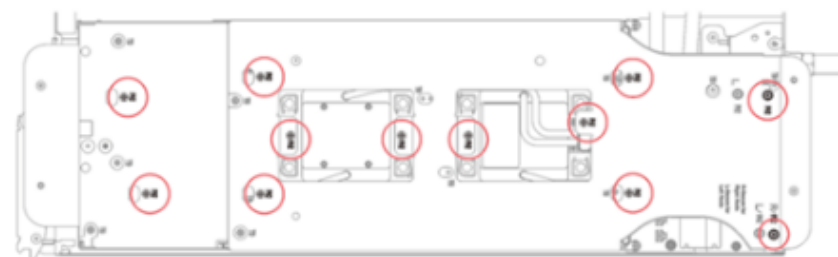
Follow the instructions in the User Guide to select the proper screwdriver to replace parts.

Remove the following screws to loosen the quick connect.

- Two **Torx T10 screws** (per node) on the quick connect.
- Five **Torx T10 screws** (per node) on the rear of the node.



Step 6. Tighten water loop carrier screws (12x **Phillips #2 screws** per node).

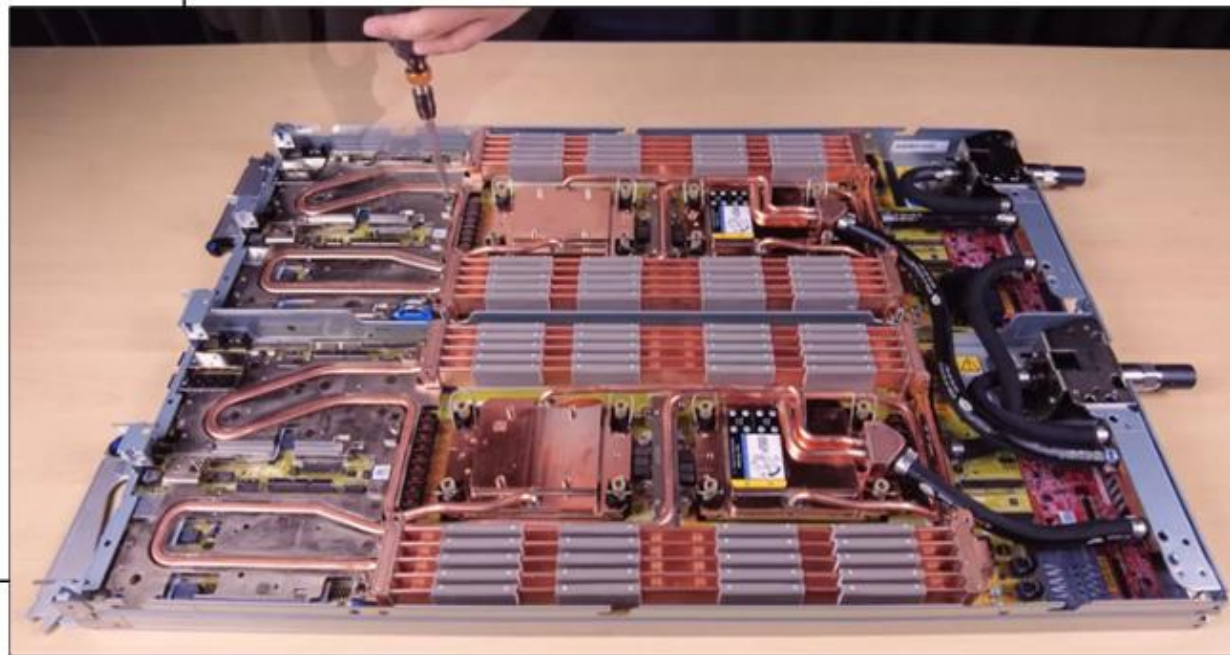
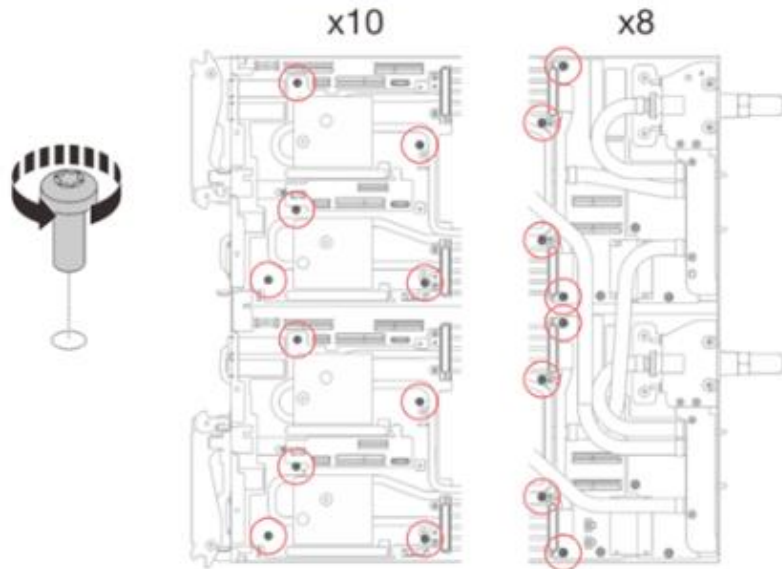


Torque screwdriver

The replacement of some SD650 V3, SD650-N V3, and SD650-I V3 parts – for example, a water loop or GPU – requires a torque screwdriver (part number: 03GY000) set to the proper setting. Refer to the User Guide for complete torque setting information for SD650 V3, SD650-N V3, and SD650-I V3 parts replacement.

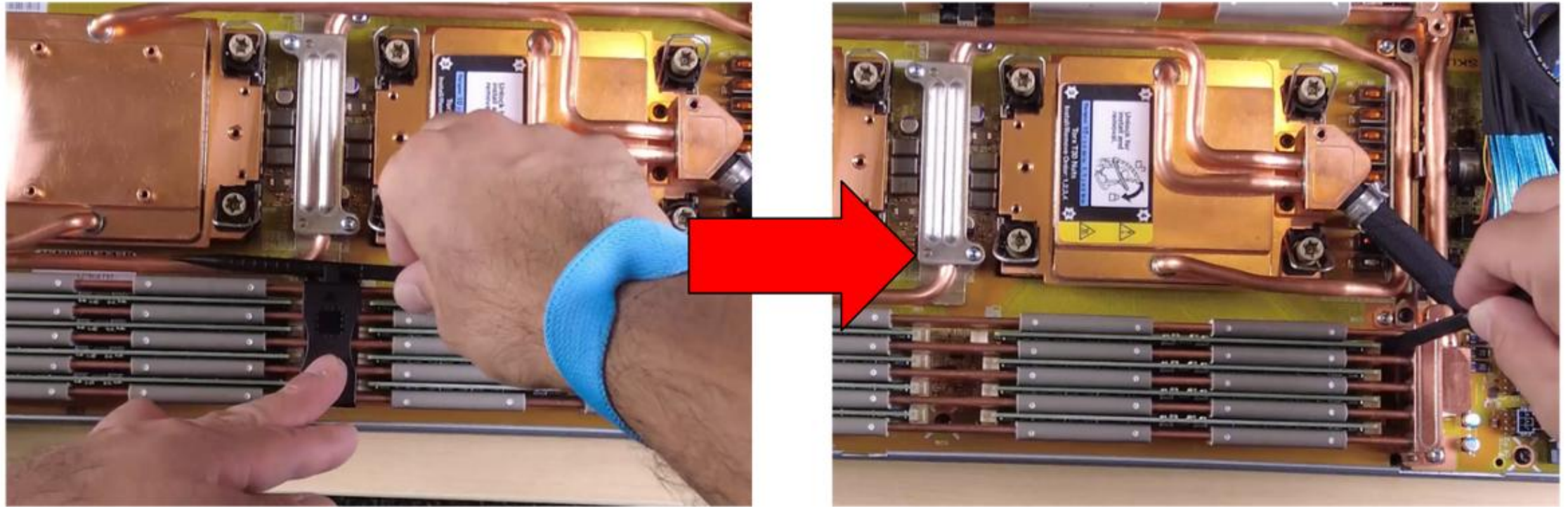
Step 4. Remove water loop screws (18x Torx T10 screws for two nodes) with a torque screwdriver set to the proper torque.

Note: For reference, the torque required for the screws to be fully tightened/removed is 0.5-0.6 newton-meters, 4.5-5.5 inch-pounds.



DIMM tool

Due to space limitations inside the SD650 V3, SD650-N V3, and SD650-I V3, a DIMM tool is recommended when opening the retaining clips to remove or install a DIMM. The DIMM tool is fitted in the CPU node.



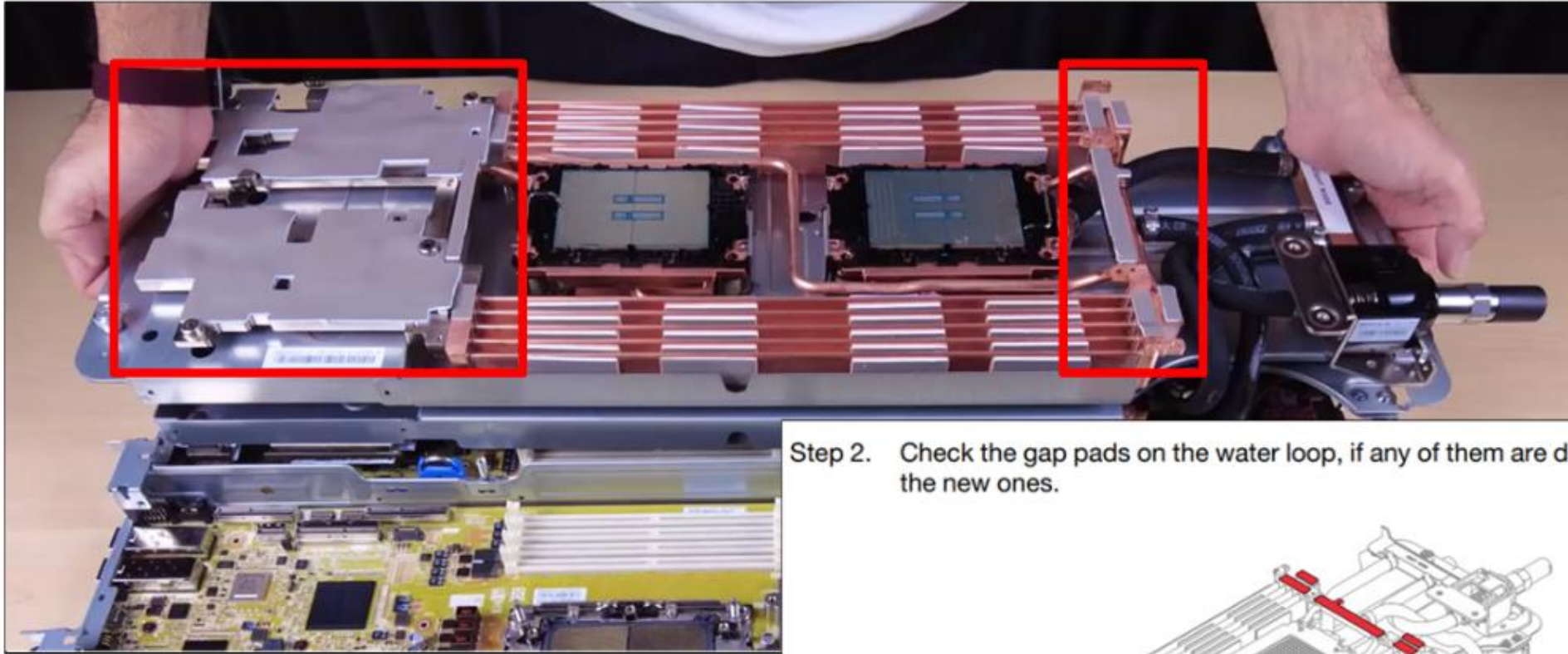
Gap pads

Many SD650 V3, SD650-N V3, and SD650-I V3 parts – for example, water loops or PCIe adapter interface plates – are covered with gap pads for thermal enhancement purposes. When replacing these parts, check the gap pads and replace any that are damaged or missing.

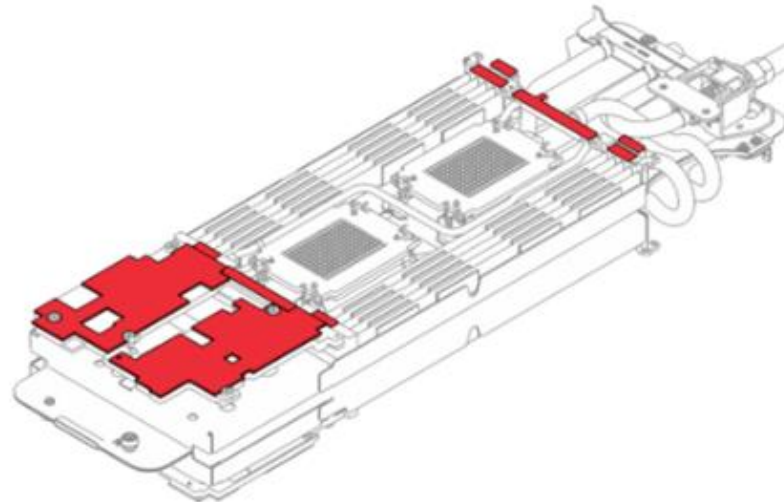
Click [HERE](#) to see an example of the location of water loop gap pads.

For complete gap pad location information, refer to the Gap pad/Putty pad identification and location section in the User Guide.

Water loop gap pads



Step 2. Check the gap pads on the water loop, if any of them are damaged or missing, replace them with the new ones.



Putty pads

Many SD650 V3, SD650-N V3, and SD650-I V3 parts – for example, water loops or M.2 adapters – are covered with putty pads for thermal enhancement purposes.

Putty pads are very fragile and can be easily damaged, so when replacing a part with putty pads, you must also replace the putty pads.

Click [HERE](#) to see an example of M.2 backplane putty pad location.

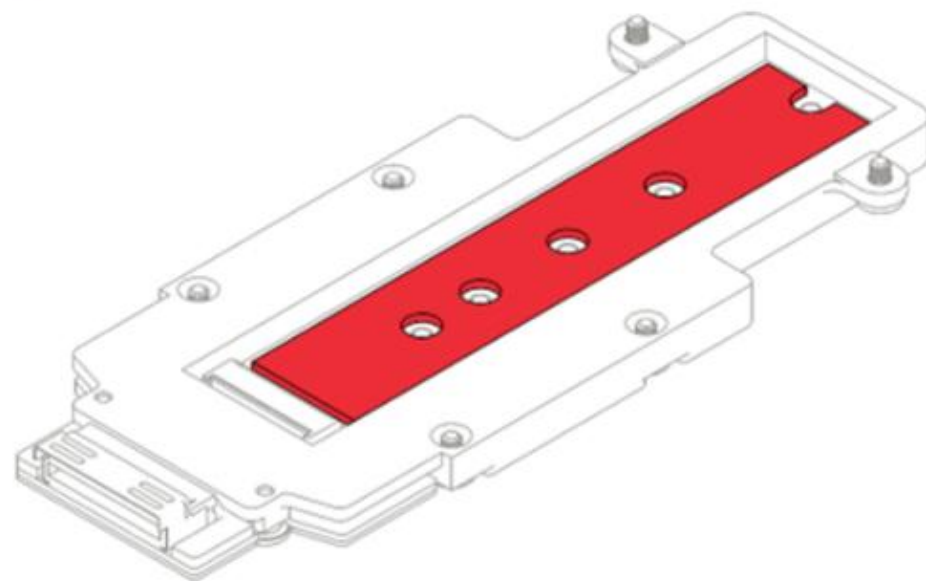
Putty pads have an expiry date, which is displayed on the packaging. Check the expiry date of new pads before you use them in a system. Do not use expired pads on any system parts.

Click [HERE](#) to see a picture of an expiry date label.

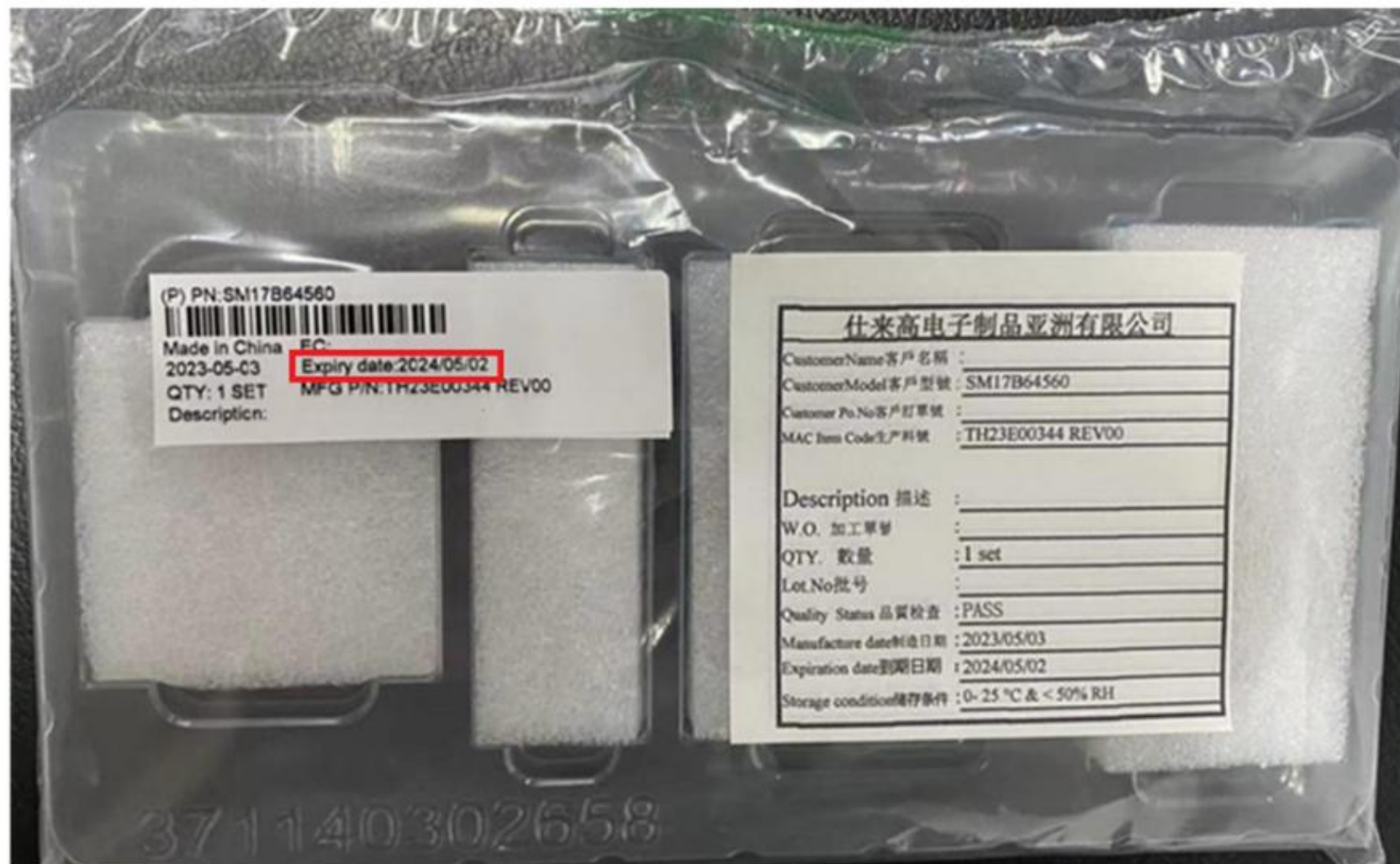
When non-expired pads have been applied to a part and the system is running, the expiry date is no longer relevant.

For complete putty pad location information, refer to the Gap pad/Putty pad identification and location section in the User Guide.

M.2 backplane putty pad



Pad expiry date label

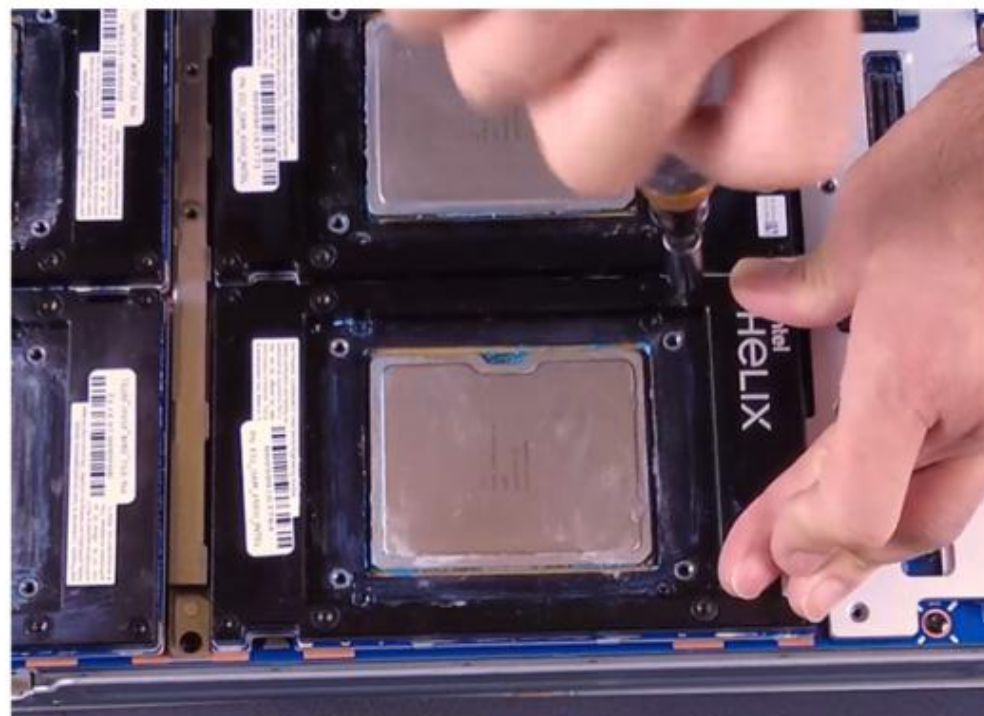
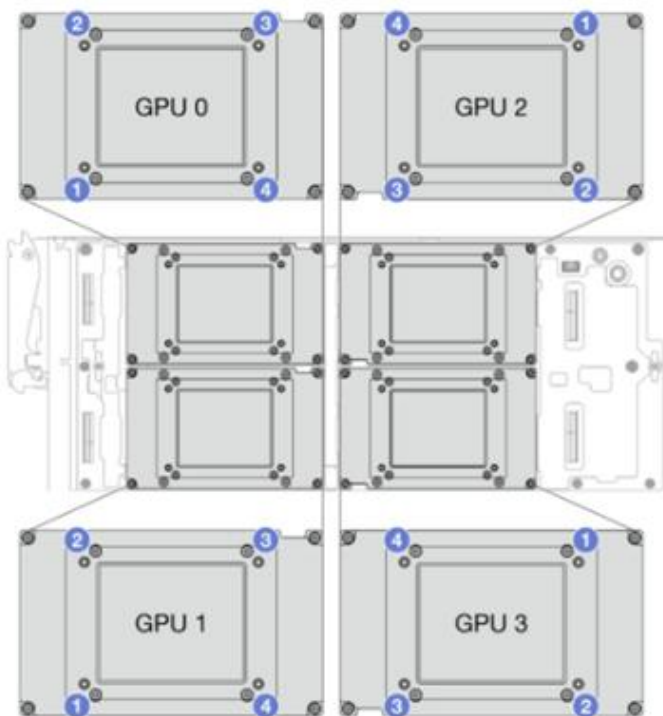
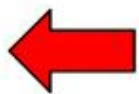


Intel GPU installation tips

When tightening the intel GPU screws, follow the sequence shown in this diagram.

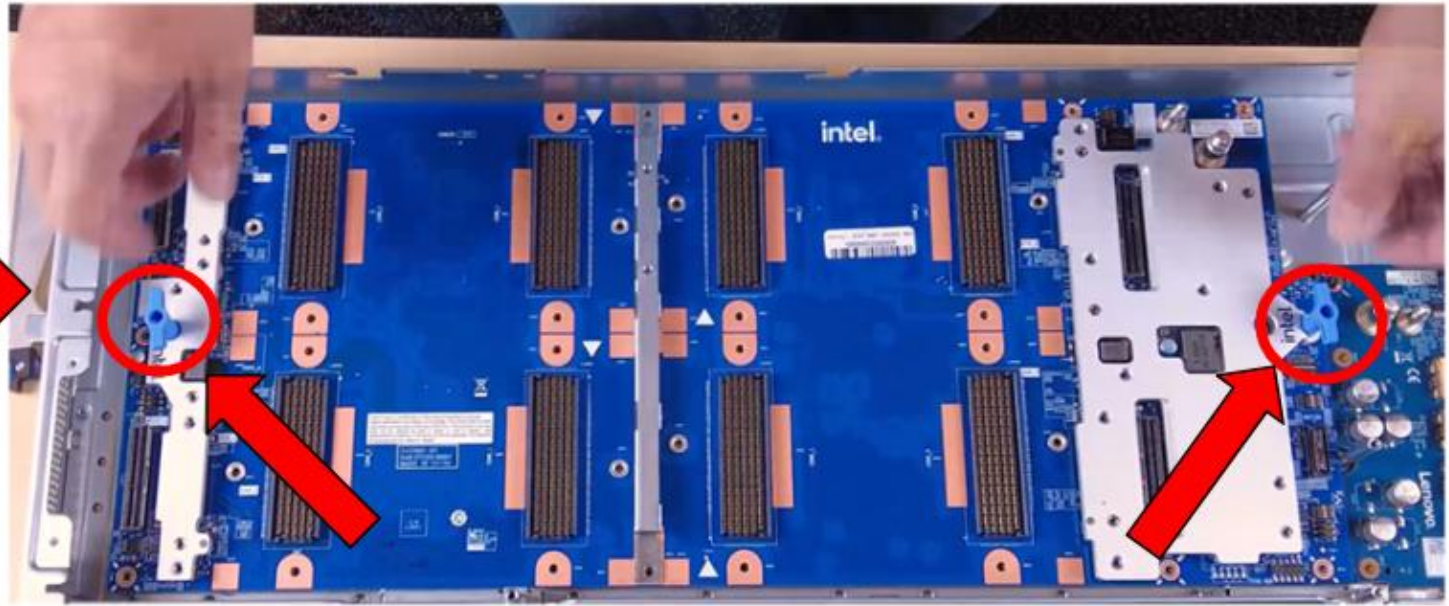
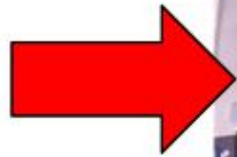
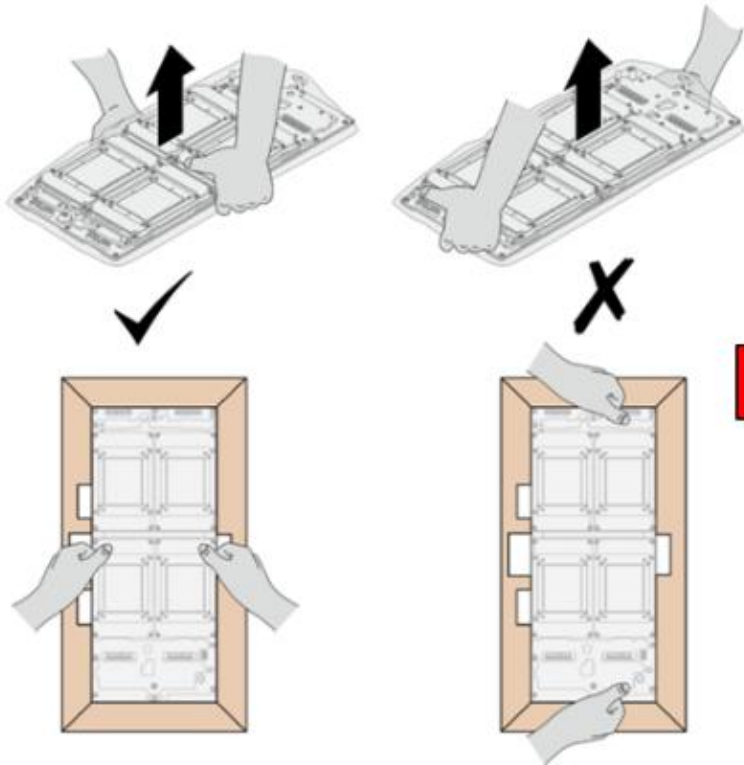
- First, set the torque screwdriver to 1 kgf.cm (0.868 lbf.in) to slightly tighten the screws.
- Then, set the torque screwdriver to 9 kgf.cm (7.8 lbf.in) to fully tighten the screws

Front of
the node



GPU board

- When the GPU board is still in the ESD bag, hold the long sides of the board. You might damage the GPU board if you hold it by the short sides.
- When replacing a GPU board, hold it by the board handles.



GPU board handles

SMM2 replacement tips

After replacing an SMM2 in a DW612S enclosure, use IPMI commands to reconfigure the SMM2 chassis ID to match the corresponding PSU configuration in the enclosure. Otherwise, SMM2 will not be able to recognize the PSUs in the enclosure and will report PSU mismatch error messages.

Work through the SMM2 OEM IPMI commands listed on the following pages to understand how to reconfigure the chassis type.



Click each step in turn to see the procedure

Step **1** — **2** — **3** — **4** — **5** — **6**

SMM2 replacement tips

By default, the SMM2 IPMI function is disabled. Use the following command to enable an SMM2 IPMI interface through XCC2:

```
ipmitool -I lanplus -H <XCC2 IP address> -U <XCC2 username> -P <XCC2 password> raw 0x3A 0xF1 0x04
```

Note: For more information about using IPMI commands in XCC, refer to the following ThinkSystem documentation: [Managing the XClarity Controller with IPMI](#).

Step **1**—**2**—**3**—**4**—**5**—**6**



SMM2 replacement tips

Use the following IPMI command to check the current enclosure chassis type:

```
ipmitool -I lanplus -H <SMM2 IP address> -U <SMM2 username> -P <SMM2 password>  
raw 0x32 0xC8 0x01
```

There are six DW612S enclosure chassis types:

Type	Descriptions	Chassis type value
Type 1	Six air-cooled PSUs only	0X01
Type 2	Six air-cooled PSUs, can be upgraded to nine PSUs	0X02
Type 3	Nine air-cooled PSUs	0X03
Type 4	Two DWC PSU modules only	0X04
Type 5	Two DWC PSU modules, can be upgraded to three DWC PSU modules	0X05
Type 6	Three DWC PSU modules	0X06

Step



SMM2 replacement tips

Use the following command to reconfigure the enclosure type:

```
ipmitool -I lanplus -H <SMM2 IP address> -U <SMM2 USERID> -P <SMM2 password> raw 0x32 0xC8 0x00 <chassis type value>
```

For example, to configure the SMM2 to DW612S enclosure type 3 (chassis type value is 0x03), use the following command:

```
ipmitool -I lanplus -H <SMM2 IP address> -U <SMM2 USERID> -P <SMM2 password> raw 0x32 0xC8 0x00 0x03
```

Step



SMM2 replacement tips

Use the following command to check the enclosure type again:

```
ipmitool -I lanplus -H [SMM2_IP] -U [USERID] -P [PASSWORD] raw 0x32  
0xC8 0x01
```

The return value should be `0x03`, which indicates that the enclosure is type 3.

Step



SMM2 replacement tips

Use the following command to restart SMM2 and make the SMM2 chassis type configuration take effect:

```
ipmitool -I lanplus -H <SMM2 IP address> -U <SMM2 username>  
-P <SMM2 password> mc reset cold
```

Step



SMM2 replacement tips

After SMM2 restarts, use the following command to disable the SMM2 IPMI interface through the XCC2 CLI:

```
ipmitool -I lanplus -H <SMM2 IP> -U <SMM2 username> -P <SMM2 password> raw 0x3A 0xF1 0x01
```

Step **1**—**2**—**3**—**4**—**5**—**6**



DWC PSU replacement

A DWC PSU module contains three DWC PSUs. If one DWC PSU fails, the servicer must replace the entire DWC PSU module, which means the enclosure will lose three PSUs during the replacement procedure. To ensure you do not impact the performance of the nodes in the enclosure, it is strongly advised that you reduce the power loading of each node installed in the enclosure before you remove the DWC PSU module. Do this by putting the nodes into the idle or power off state.

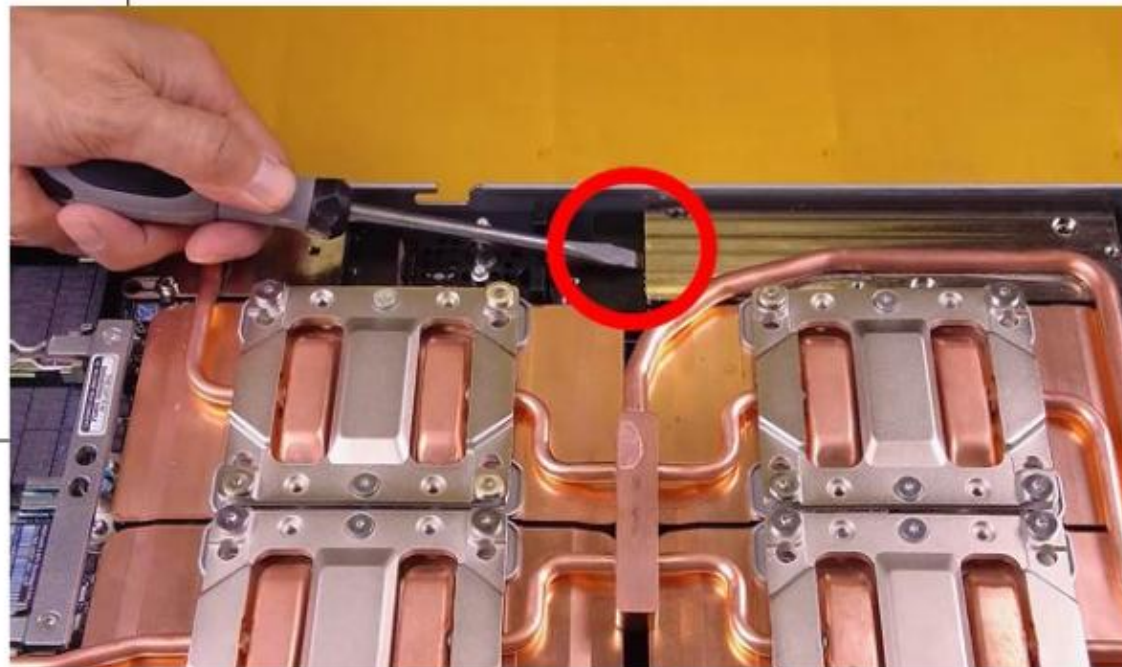
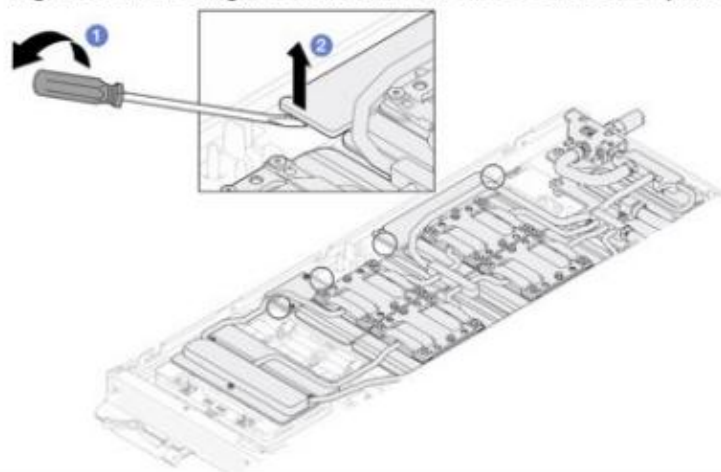


GPU node water loop replacement tip

When removing a water loop from the GPU node, you might need to use a flat-head screwdriver to separate specific components from the GPU board. Refer to the User Guide for more information.

b. ② The front and rear MISC conduction plates release from the GPU board slightly.

Figure 8. Releasing the front and rear MISC conduction plates release from the GPU board



NVIDIA GPU PCM replacement

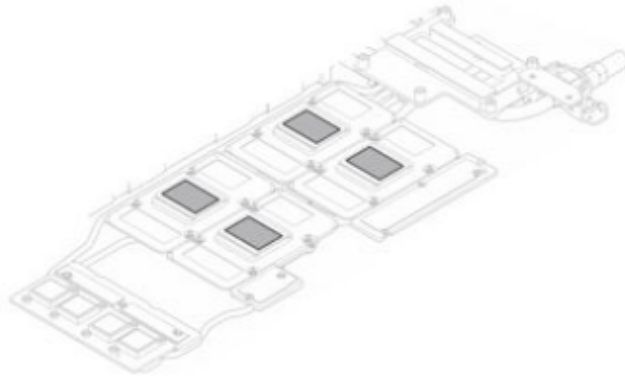
NVIDIA GPUs use a specific thermal material called PCM. After removing a water loop from the NVIDIA GPU side, it is recommended that you immediately clean off the PCM while it is in a liquid state. Dry PCM is difficult to remove, and attempting to do so might cause you to damage the GPU.

14. **Immediately** clean the PCM off all the GPUs with alcohol cleaning pads. **Gently** clean the PCM to avoid GPU damages.

⚠ ATTENTION

- It is recommended to clean the PCM while it is in liquid state.
- The electrical components around the die on the GPUs are extremely delicate. When removing the PCM and cleaning the GPU die, avoid touching the electrical components to prevent damages.

Figure 14. Cleaning PCM off from all GPUs



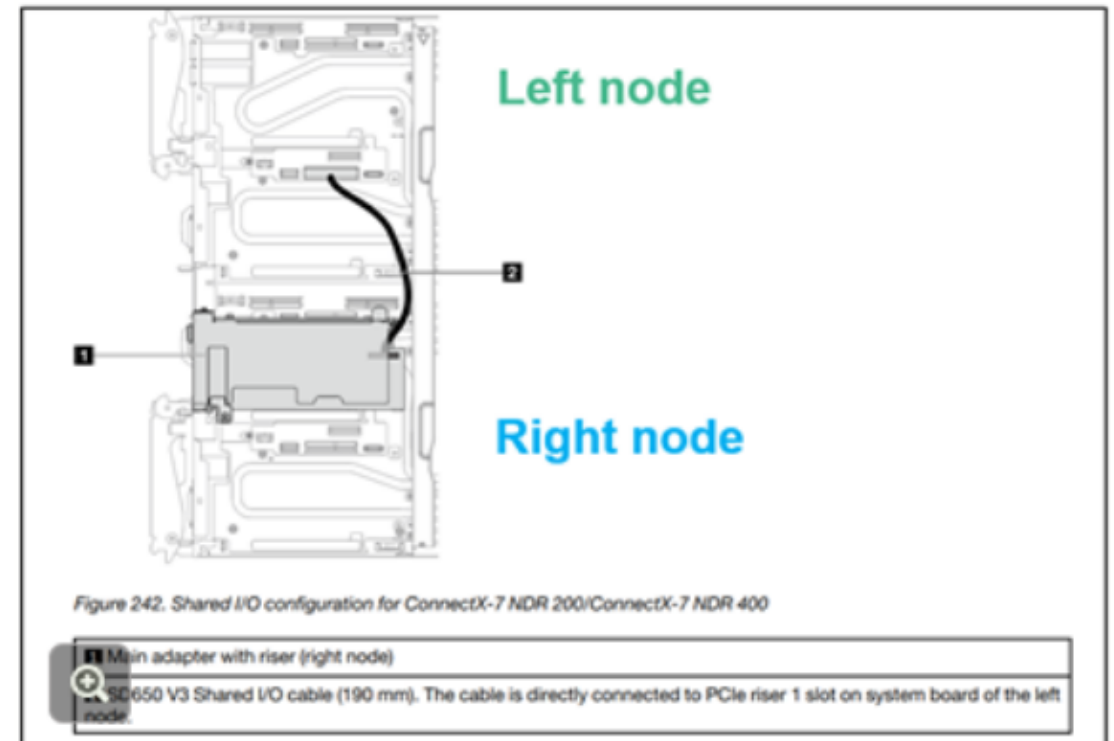
Powering on or off an SD650 V3 with a shared I/O configuration

The two nodes of SD650 V3 systems with a shared I/O configuration require specific power on and power off sequences:

- Power on sequence: Power on the node with the main adapter (right node), and then power on the left node
- Power off sequence: Power off the left node, and then power off the node with the main adapter (right node)



An SD650 V3 with a shared I/O configuration for the ConnectX-7 NDR 400 adapter

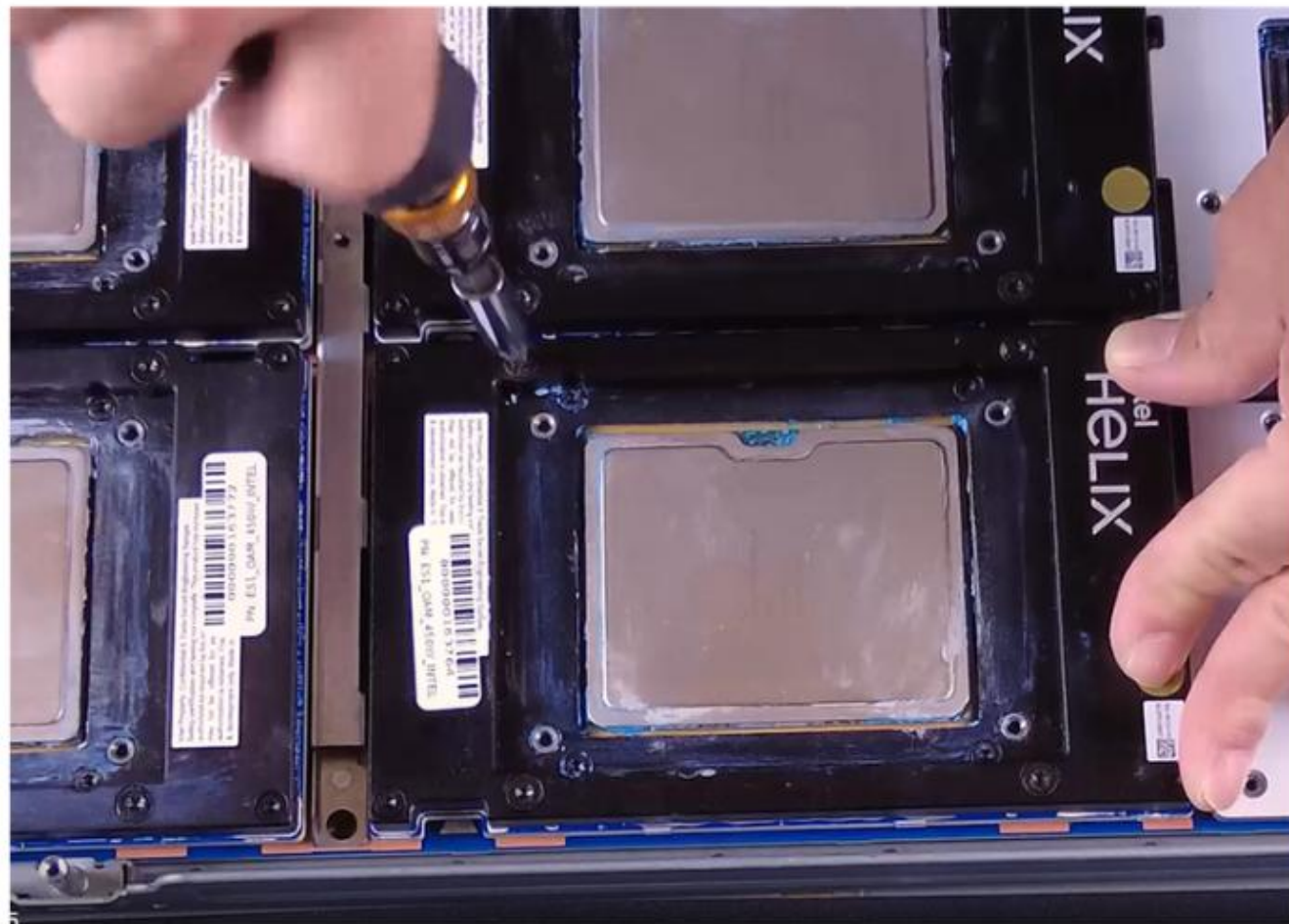


Intel GPU OAM replacement overview

After replacing an Intel GPU OAM in an SD650-I V3, servicers must work through the following steps to update the firmware packages:

1. Use XCC to update the Intel AMC firmware.
2. Use the [Intel XPU Manager command line tool \(XPUM\)](#) to update the Intel IFWI firmware in the operating system.

The following slides describe these steps in more detail.



Updating the Intel AMC firmware

To update the Intel AMC firmware, go to Lenovo Support to download the AMC firmware, and then go to the XCC Firmware Update page to update the AMC firmware. Make sure all four Intel GPU OAMs are running the same AMC firmware version.

The screenshot displays the XCC Firmware Update interface. On the left, a sidebar contains navigation options: Remote Console, Firmware Update (highlighted with a red box), Storage, Server Configuration, and BMC Configuration. The main content area is titled 'Adapter Firmware' and includes a description: 'Update adapter firmware with granular selection of an individual adapter or multiple adapters of the same or different types, depending on the payload content.' A red box highlights the 'Update Firmware' button in the top right corner of the main content area. Below this, a note states: 'Note: the system must have completed booting at least once for all adapters to be detected. Activation of retimer device'. A red arrow points from this note to a modal window titled 'Verify Firmware'. The modal window has three steps: 1. Select File, 2. Verify Firmware (current step), and 3. Update Firmware. It shows a progress bar and a message: 'Comparison Finished. 4 firmware(s) is available. Select the firmware(s) and then click Next to start update.' Below the message is a table with the following data:

Slot No.	Device Name	New Version	Installed Version	Reboot Required
☒ 3	Ponte Vecchio	6.0.5.0	6.0.0.0	Yes
☒ 4	Ponte Vecchio	6.0.5.0	6.0.0.0	Yes
☒ 5	Ponte Vecchio	6.0.5.0	6.0.0.0	Yes
☒ 6	Ponte Vecchio	6.0.5.0	6.0.0.0	Yes

At the bottom of the modal window, there are buttons for '< Back', 'Next >', and 'Cancel'.

Updating the Intel IFWI firmware

After replacing an Intel GPU OAM, servicers must use the [Intel XPU Manager command line tool \(XPUM\)](#) to update the Intel IFWI firmware in the operating system.

In this example, Intel IFWI firmware in an RHEL 8.6 OS environment will be updated.



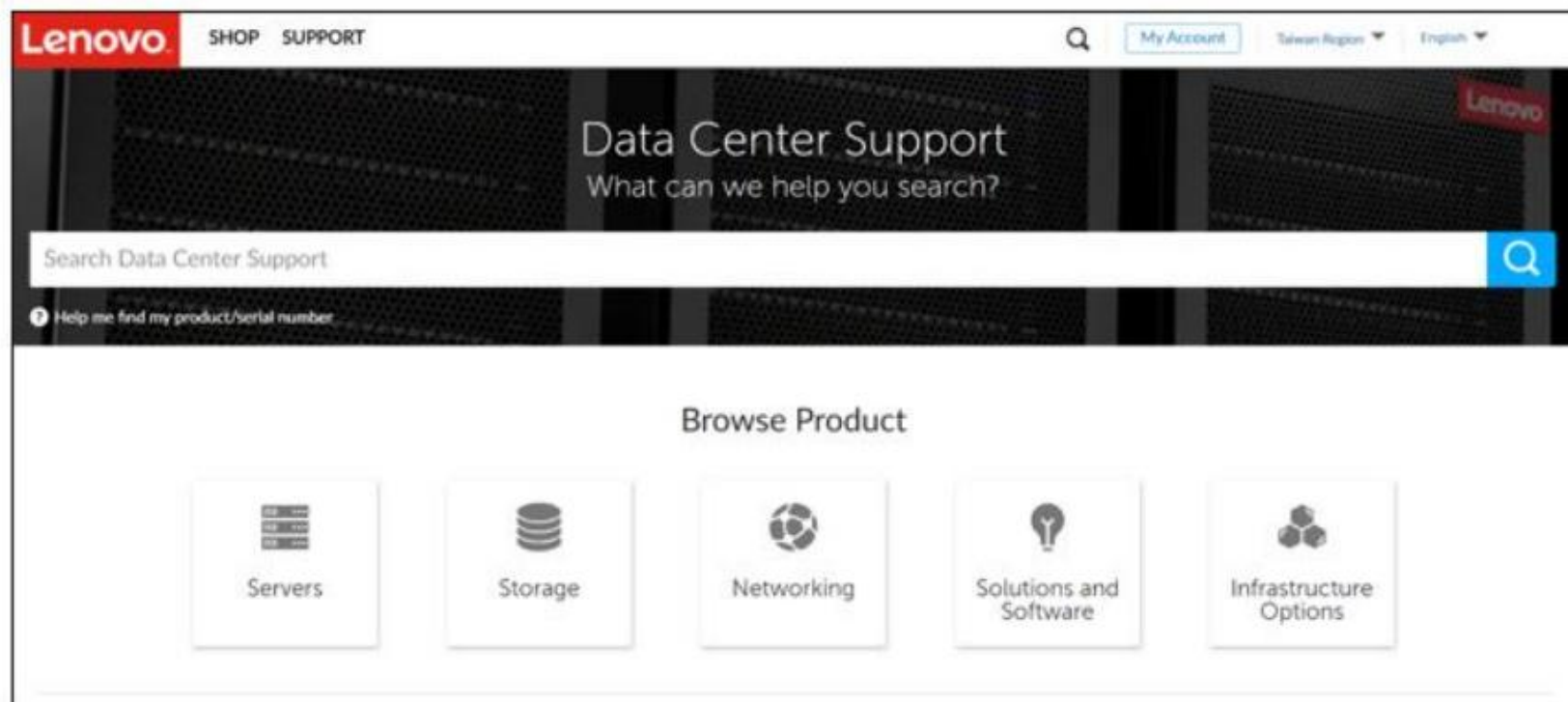
Click each step in turn to see the procedure

Step



Updating the Intel IFWI firmware

Go to [Lenovo Support](#) to download the Intel IFWI firmware.

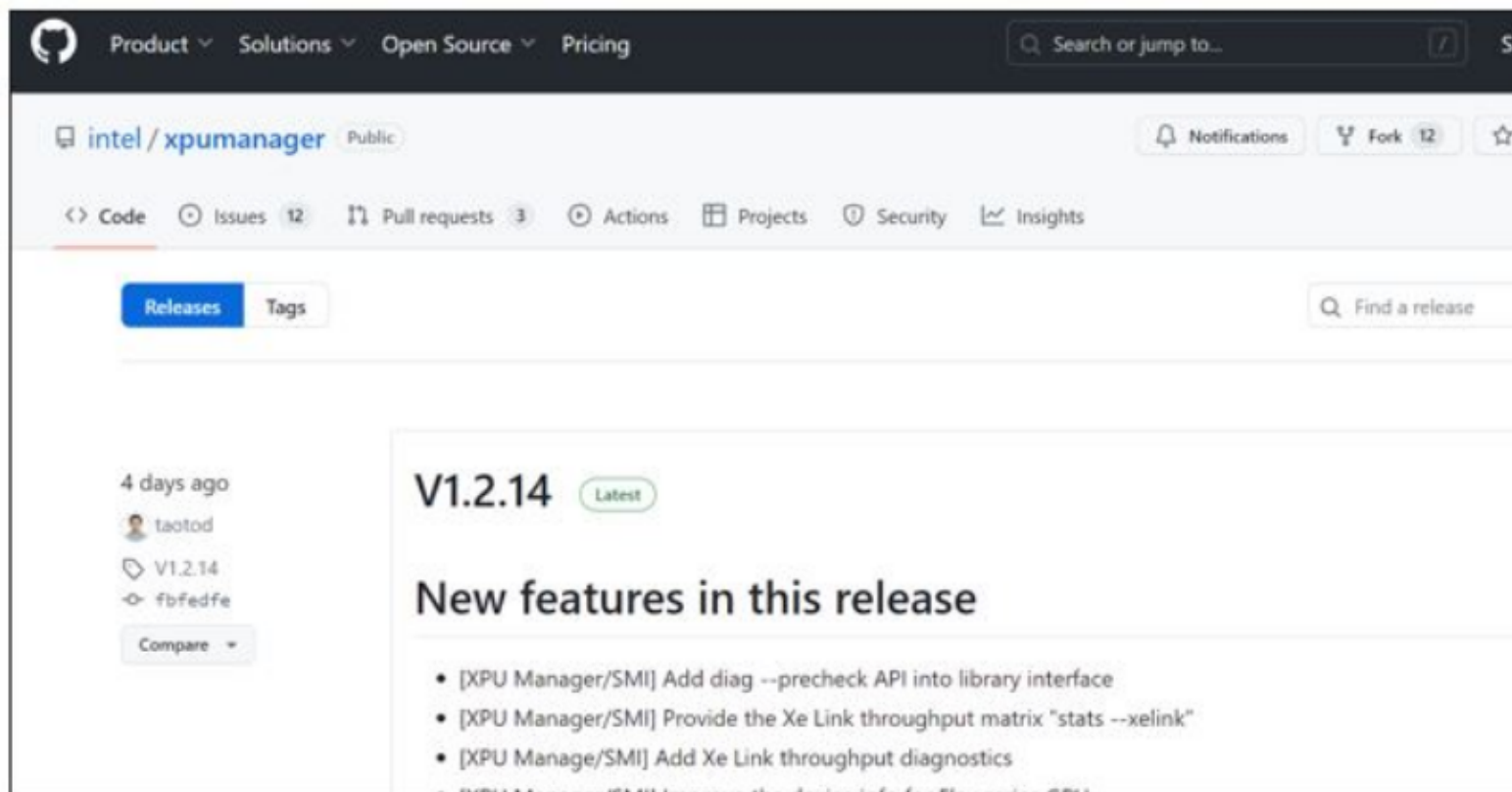


Step



Updating the Intel IFWI firmware

Download the Intel XPUM tool from the [Intel GitHub](#) web page.



Step



Updating the Intel IFWI firmware

On the [XPUM Releases page](#), check the **Supported OSes** section for the supported OS information, and then go to the **Assets** section to download the corresponding installer files. For example, to download the installer for RHEL 8.5 or 8.6, click the download link with *.el8.x86_64.rpm in the file name.

The screenshot displays two panels from the XPUM Releases page. The left panel, titled 'Supported OSes', lists supported operating systems and their corresponding installer files. The right panel, titled 'Assets', shows a list of 20 assets with their file names, sizes, and upload dates.

Supported OSes

- XPU-SMI
 - CentOS 8/9 Stream (installer files: xpu-smi-*el8/el9.x86_64.rpm)
 - CentOS 7.4/7.9 (installer files: xpu-smi-*el7.x86_64.rpm)
 - Ubuntu 20.04.3/22.04 (installer files: xpu-smi-*u22.04/22.04_amd64.rpm)
 - RHEL 8.5/8.6 (installer files: xpu-smi-*el8.x86_64.rpm)
 - SLES 15 SP3/SP4 (installer files: xpu-smi-*x86_64.rpm)
 - Debian 10.13 (installer files: xpu-smi-*deb10u1_amd64.deb)
 - Windows Server 2022/2019 (file: xpu-smi-*_win.zip, limited supporting: GPU device info, GPU telemetries, C settings, GPU firmware update)
- XPU Manager
 - Ubuntu 20.04.3/22.04 (installer files: xpumanager-*u20.04/u22.04_amd64.deb)
 - RHEL 8.5/8.6 (installer files: xpumanager-*el8.x86_64.rpm)
 - CentOS 8/9 Stream (installer files: xpumanager-*el8/el9.x86_64.rpm)
 - CentOS 7.4/7.9 (installer files: xpumanager-*el7.x86_64.rpm)
 - SLES 15 SP3/SP4 (installer files: xpumanager-*x86_64.rpm)

Assets

Asset Name	Size	Time
amcncs	663 KB	4 days ago
amcncs.sig	543 Bytes	4 days ago
xpu-smi-1.2.14-20230714.022616.1881d85b_win.zip	569 KB	4 days ago
xpu-smi-1.2.14-20230714.022616.1881d85b_win.zip.sig	9.79 KB	4 days ago
xpu-smi-1.2.14-20230714.022726.1881d85b.el7.x86_64.rpm	20.7 MB	4 days ago
xpu-smi-1.2.14-20230714.022726.1881d85b.el8.x86_64.rpm	20.6 MB	4 days ago
xpu-smi-1.2.14-20230714.022726.1881d85b.el9.x86_64.rpm	20.7 MB	4 days ago
xpu-smi-1.2.14-20230714.022726.1881d85b.x86_64.rpm	20.5 MB	4 days ago
xpu-smi-1.2.14-20230714.022726.1881d85b-deb10u1_amd64.deb	22.2 MB	4 days ago
xpu-smi-1.2.14-20230714.022726.1881d85b.u20.04_amd64.deb	22.3 MB	4 days ago
Source code (zip)		4 days ago
Source code (tar.gz)		4 days ago

Show all 20 assets

Step



Updating the Intel IFWI firmware

After downloading the installation files, use the `sudo rpm -ivh xpumanager <XPUM installer file name>` command to install the XPUM tool in RHEL 8.6.

```
[root@localhost Desktop]# rpm -ivh xpumanager-1.2.12-20230615.060233.005a974f.el8.x86_64.rpm
warning: xpumanager-1.2.12-20230615.060233.005a974f.el8.x86_64.rpm: Header V4 RS
A/SHA256 Signature, key ID a0661990: NOKEY
Verifying...                               ##### [100%]
Preparing...                               ##### [100%]
Updating / installing...
 1:xpumanager-1.2.12-20230615.060233##### [100%]
```

Step



Updating the Intel IFWI firmware

After installing the XPUM, use the `#sudo xpumcli discovery` command to see information about all the GPU OAMs installed on the system.

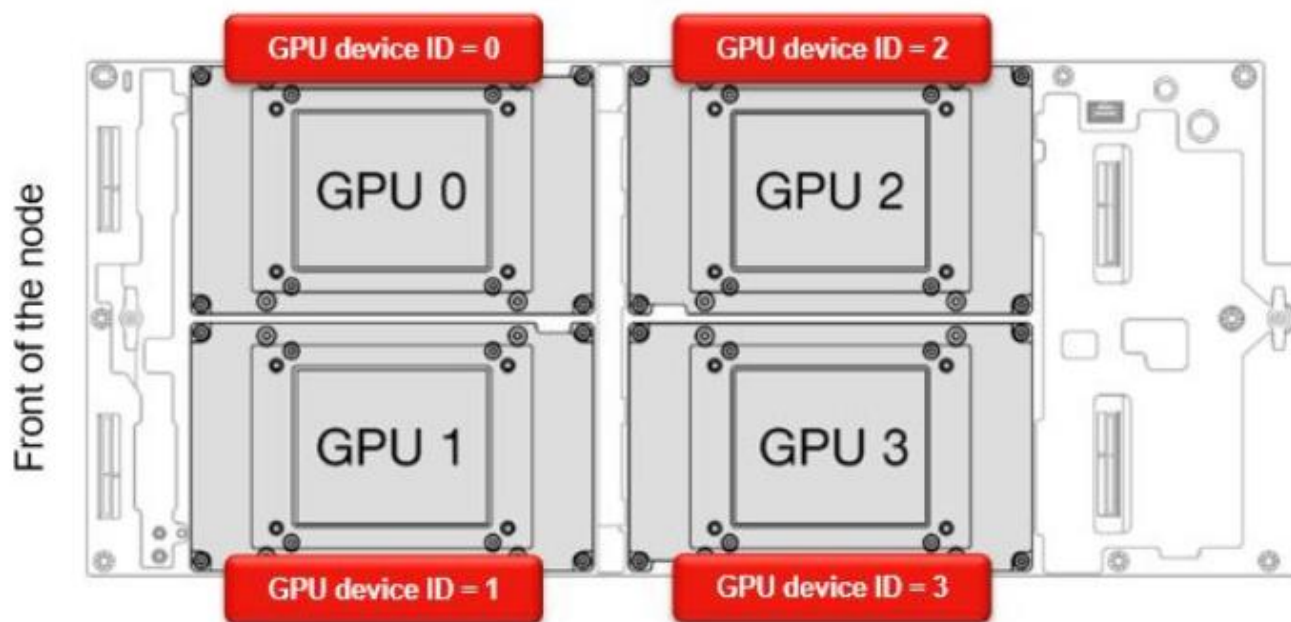
```
[root@localhost ~]# xpumcli discovery
+-----+-----+
+-----+
| Device ID | Device Information
|           |
+-----+-----+
+-----+
| 0         | Device Name: Intel Corporation Device 56c1 (rev 05)
|           |
|           | Vendor Name: Intel(R) Corporation
|           |
```

Step

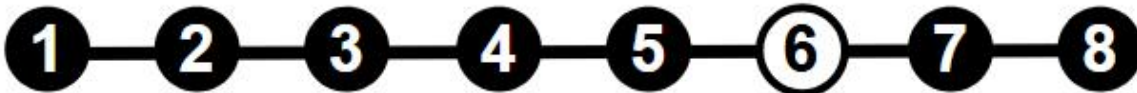


Updating the Intel IFWI firmware

Use the `#xpumcli updatefw -d <device ID> -t GFX -f <IFWI firmware file route and file name>` command to update the IFWI firmware, where *device ID* = GPU numbering. Make sure all four Intel GPU OAMs run the same IFWI firmware version.



Step

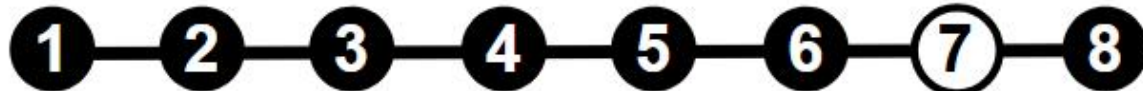


Updating the Intel IFWI firmware

For example, if you need to update the Intel IFWI firmware on GPU 0 (device ID = 0), use the `#xpumcli updatefw -d 0 -t GFX -f <IFWI firmware file route and file name>` command.

```
[root@localhost ~]# xpumcli updatefw -d 0 -t GFX -f /home/PVC.Fwupdate_Prod_2023.WW23.3_Tuscany.bin
This GPU card has multiple cores. This operation will update all firmwares. Do you want to continue? (y/n) y
Device 0 FW version: DG02_2.2314
Device 1 FW version: DG02_2.2314
Image FW version: DG02_2.2273
Do you want to continue? (y/n) y
Start to update firmware
Firmware Name: GFX
Image path: /home/PVC.Fwupdate_Prod_2023.WW23.3_Tuscany.bin
[=====] 100 %
Update firmware successfully.
[root@localhost ~]#
```

Step



Updating the Intel IFWI firmware

For more information about XPU CLI commands, refer to the Intel(R) XPU Manager Command Line Interface User Guide on the [Intel GitHub](#) website.

Intel(R) XPU Manager Command Line Interface User Guide

This guide describes how to use XPU Manager Command Line Interface to manage Intel GPU devices.

Intel(R) XPU Manager Command Line Interface main features

- Show the device info.
- Manage multiple devices by the group-level.
- Get lots of raw and aggregated device statistics.
- Get the health status of the device components

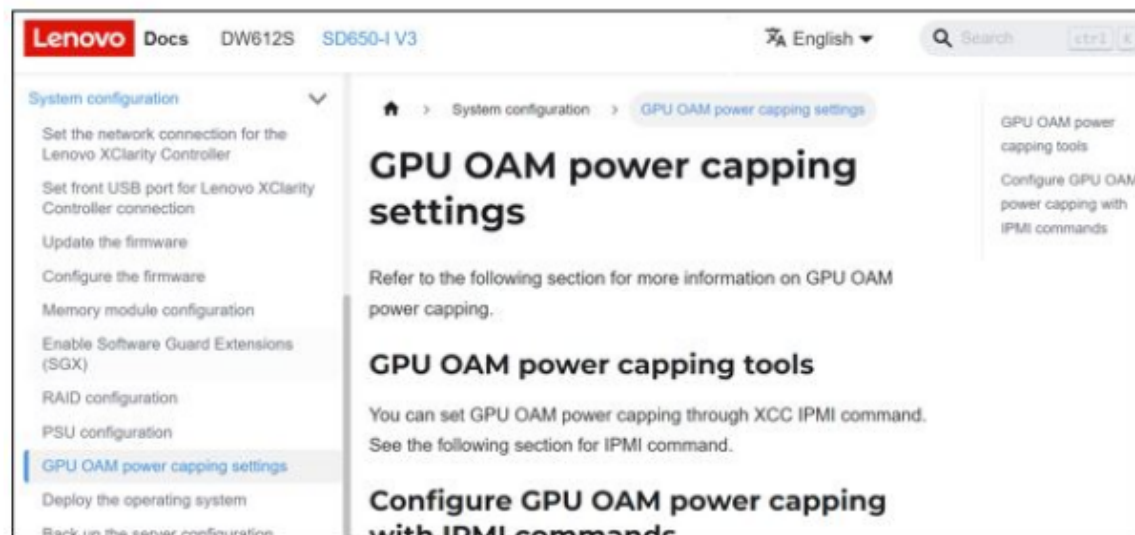
Step



Adjusting power capping settings

Before replacing a system board, use IPMI commands to check the current GPU OAM power capping settings.

After replacing a system board, if necessary, use IPMI commands to check the GPU OAM power capping setting matches the customer environment setting. For more information about adjusting the GPU OAM power capping setting, refer to the *GPU OAM power capping settings* page of the *System configuration* section of the *SD650-I V3 user guide* on [Lenovo Docs](#).



2. Set power capping with command line below:

```
ipmitool raw 0x3a 0xce 9 3 0x01 0x0[x] 0x[y][z]
```

Where[x], [y], and [z] are the first, second, and third digits respectively of the converted hexadecimal number.

For example, the command line for power capping GPU to 580W is:

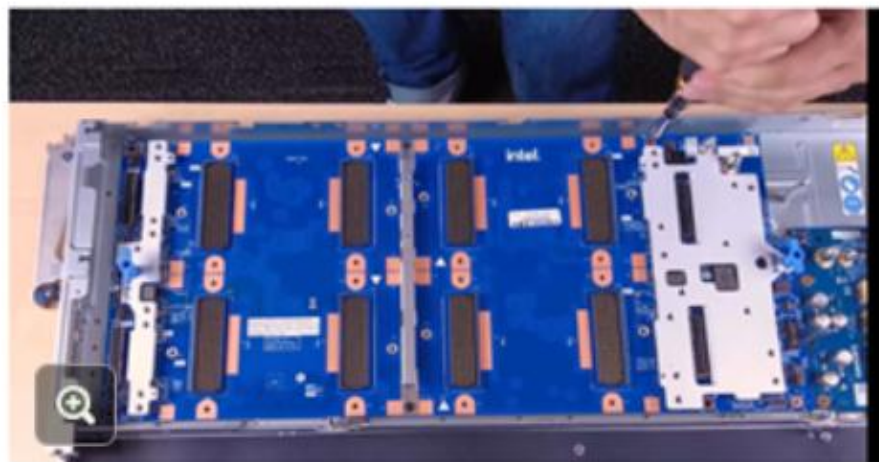
```
ipmitool raw 0x3a 0xce 9 3 0x01 0x02 0x44
```


Replacing an Intel GPU carrier base board

After replacing an Intel GPU carrier base board (CBB) in an SD650-I V3, the servicer must update the following firmware code:

- Retimer firmware (The retimer is embedded in the CBB)
- XCC firmware
- Go to SMM2 to perform a virtual reseal on the tray.

Because CBB's FPGA code is stored in the system board's XCC firmware, not CBB itself. After replacing a CBB, to make sure the CBB FPGA code is updated to the latest level, servicer must update the XCC firmware (even if the XCC's firmware is already running the latest version) for the latest CBB FPGA code to take effect.



Summary

This course enabled you to:

- Describe the features and specifications of the ThinkSystem SD650 V3 server, SD650-I V3 GPU node tray, SD650-N V3 node tray, and DW612S enclosure
- Identify the ThinkSystem SD650 V3 server, SD650-I V3 GPU node tray, SD650-N V3 node tray, and DW612S enclosure unit and components
- Describe the features and functions of SMM2
- Describe the specific problem determination steps and explain how to troubleshoot issues with the SD650 V3 server, SD650-I V3 GPU node tray, SD650-N V3 node tray, and DW612S enclosure unit